# Internship Task 1: Splunk Log Analysis

**Internship Domain:** Cybersecurity — Security Operations Center (SOC)
**Tool Used:** Splunk Cloud
**Task:** Log Analysis and Monitoring

## 1. Objective

The objective of this task was to learn how to upload log data into Splunk, search events, identify error responses, analyze IP addresses, and examine HTTP status codes.

## 2. Tools and Technologies

Splunk Cloud

Search & Reporting

Search Processing Language (SPL)

Sample web server log file (sample_web_logs.log)

## 3. Procedure

### Step 1: Data Upload

I uploaded a sample web server log file named sample_web_logs.log to Splunk Cloud. The sourcetype was set to access_combined, and the data was indexed successfully.

### Step 2: Viewing Log Events

I used the following query to view the indexed events:

```
index=main
```

Splunk displayed 20 events. The sample file contained 10 log lines, but the events appeared duplicated in Splunk. This was noted as a data-quality issue during analysis.

### Step 3: Error Analysis

I used this query to identify HTTP error responses:

```
index=main sourcetype=access_combined
| where status>=400
```

The search returned 14 events, including HTTP status codes 401, 403, 404, and 500. These results included duplicated events.

### Step 4: IP Address Analysis

I used the following query to count events by client IP address:

```
index=main sourcetype=access_combined
| rex field=_raw "^(?<client_ip>\d{1,3}(?:\.\d{1,3}){3})"
| stats count by client_ip
| sort - count
```

The results showed that 203.0.113.45 appeared most frequently, with 6 events. Since the events were duplicated, these counts represent indexed events rather than unique requests.

### Step 5: HTTP Status Code Analysis

I used the following query to count events by HTTP status code:

```
index=main sourcetype=access_combined
| stats count by status
| sort - count
```

The results were:

| Status code | Meaning | Event count |
|---|---|---|
| 200 | Successful request | 6 |
| 401 | Unauthorized | 4 |
| 403 | Forbidden | 4 |
| 404 | Page not found | 4 |
| 500 | Internal server error | 2 |

These counts include duplicated events from the sample data.

# 4. Findings

1. Splunk successfully indexed the sample web logs.
2. HTTP 401 and 403 responses indicated authentication or access-control issues.
3. HTTP 404 responses indicated requests for pages that were not found.
4. HTTP 500 responses indicated server-side errors.
5. The IP address 203.0.113.45 had the highest event count in the sample.
6. Duplicate events were observed and should be investigated before drawing conclusions from real-world logs.

**Note:** The log file contained simulated practice data. The results do not confirm an actual cyberattack.

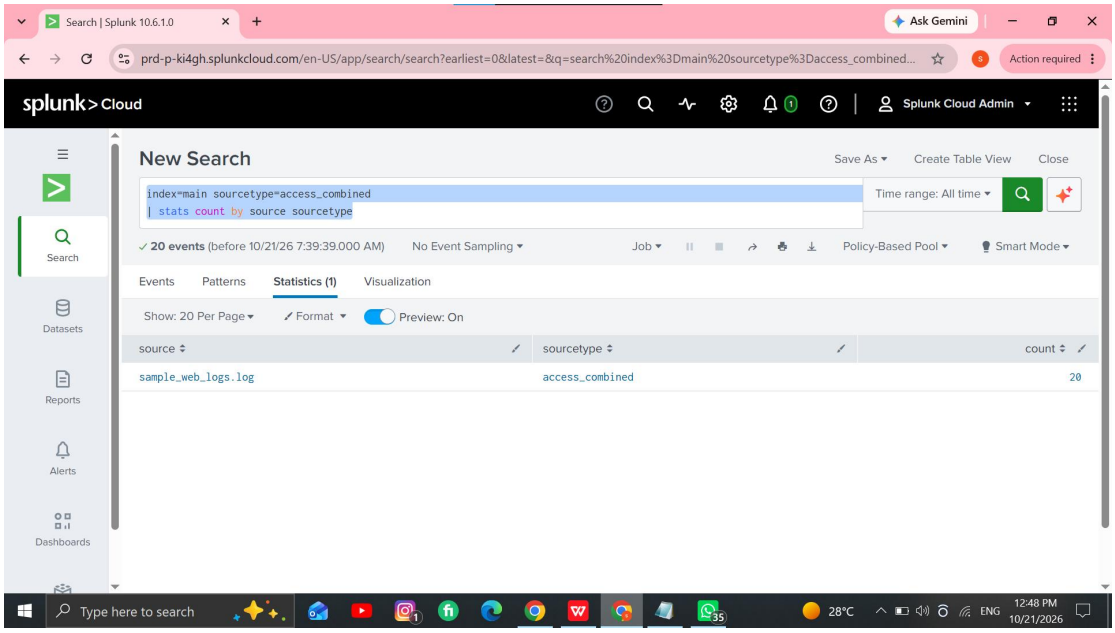
# 5. Conclusion

This task helped me learn the basics of Splunk Cloud, log searching, SPL queries, IP address analysis, and HTTP error investigation. I also learned that analysts should check data quality and validate findings before reporting a potential security incident.

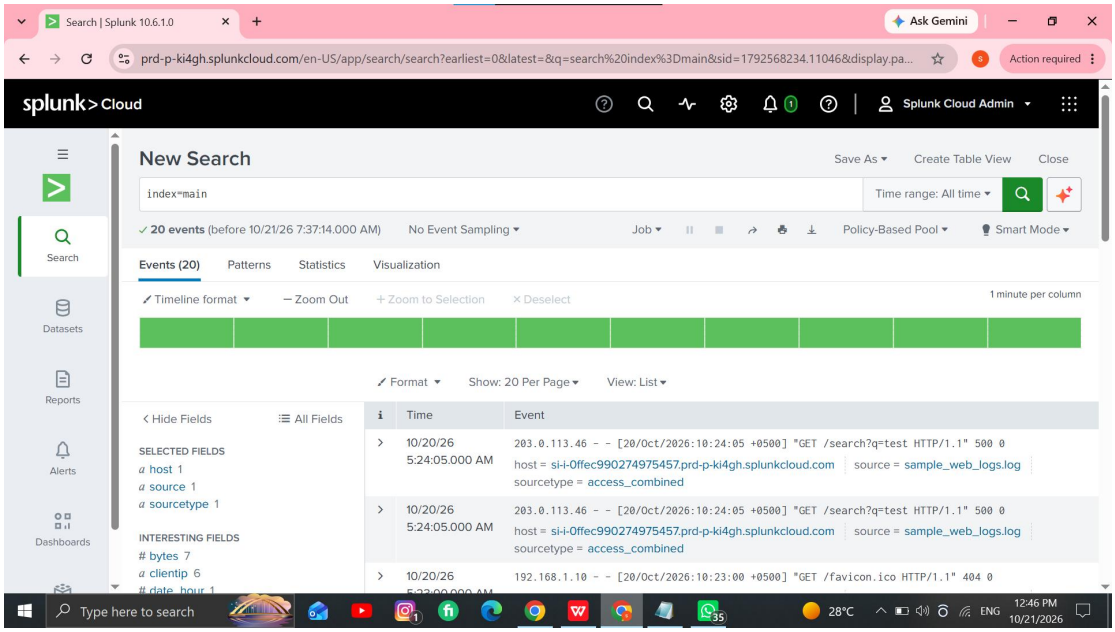
# 6. Evidence and Screenshots

Insert the following screenshots in the report:

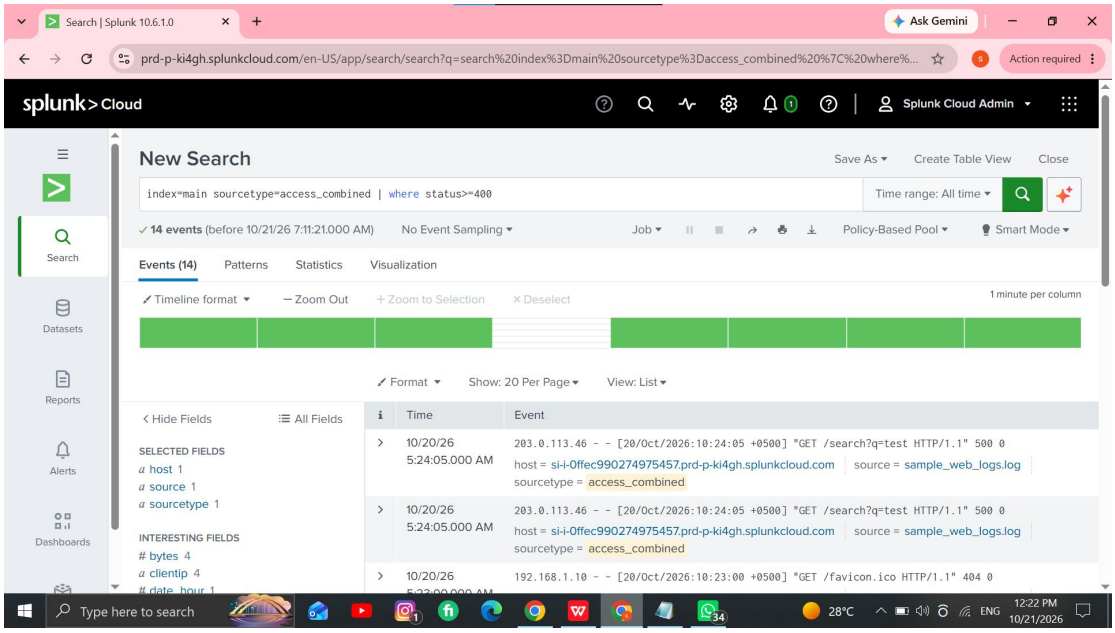
01_Data_Upload.png — Evidence showing the uploaded log file indexed in Splunk, including its source and sourcetype.



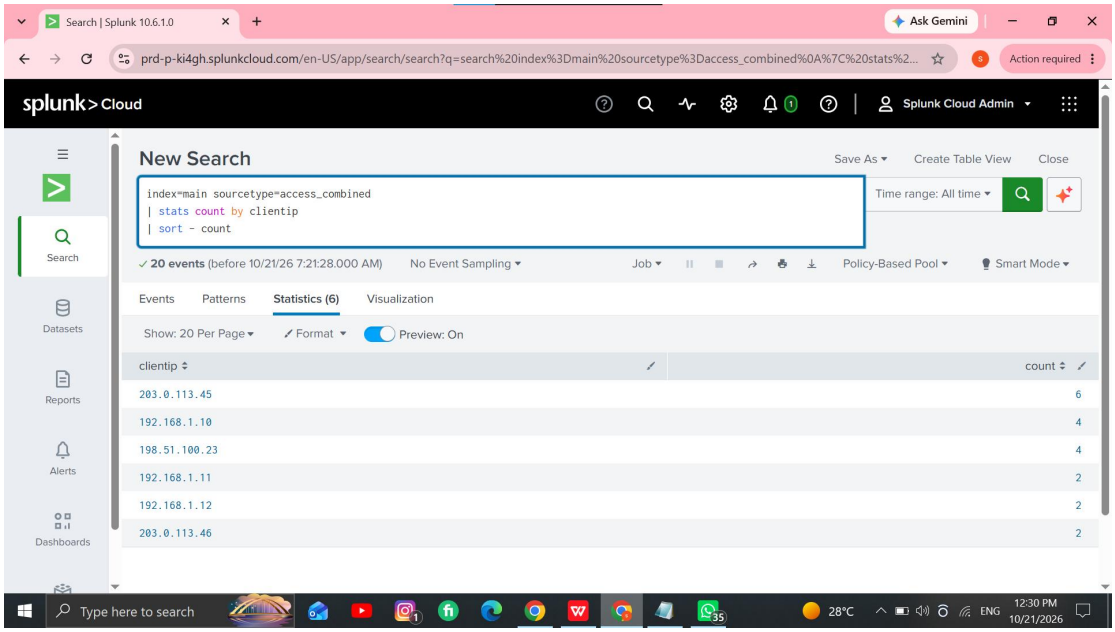
02_Initial_Logs.png — Initial log events.



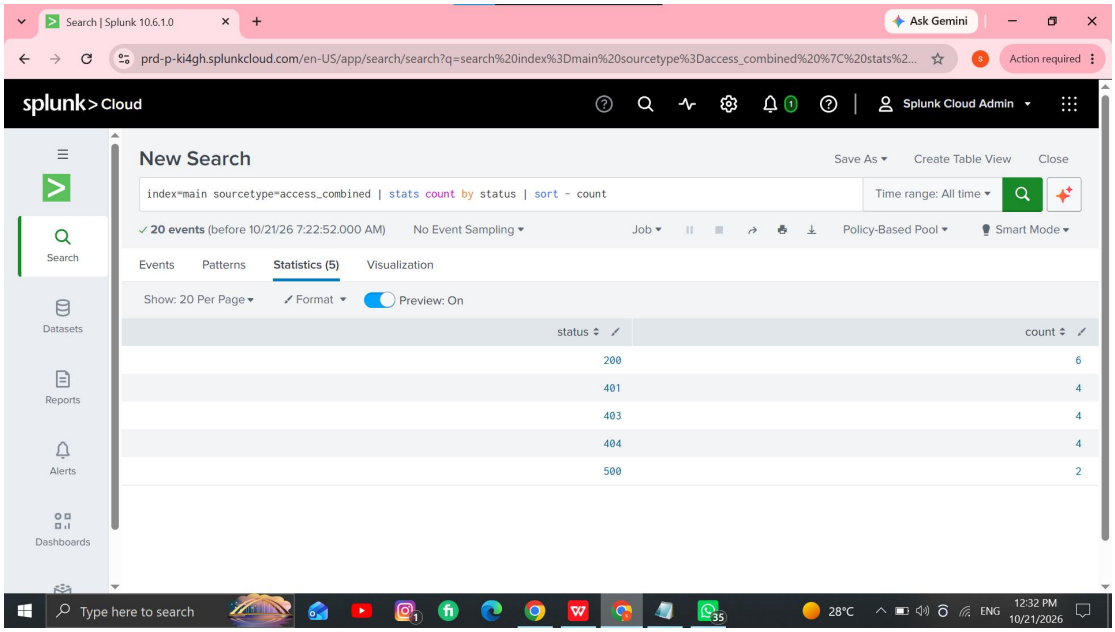

03_Error_Analysis.png — HTTP error analysis.



04_IP_Analysis.png — Event counts by IP address.




05_Status_Code_Analysis.png — Event counts by HTTP status code.



## 7. AI-Assisted SPL Query Understanding

AI can help security analysts convert plain-English requests into Splunk Search Processing Language (SPL) queries.

**Example plain-English request:** Find HTTP errors with status codes greater than or equal to 400.

**SPL query:**

```
index=main sourcetype=access_combined
| where status>=400
```

This query filters events with HTTP status codes of 400 or higher, helping analysts investigate failed requests and server errors.

AI-generated queries should always be checked and tested before being used in a real security environment.